

CSA1708-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 4 : Python Program for Crypt-Arithmetic Problem

Aim

To develop a Python program that solves the crypt-arithmetic puzzle $SEND + MORE = MONEY$ by assigning a unique digit to each letter.

Algorithm

- List the distinct letters in the puzzle: S, E, N, D, M, O, R, Y.
- Generate every possible assignment of different digits (0–9) to these letters using permutations.
- Reject any assignment in which S or M is 0, since they are leading digits.
- Compute the numeric values of SEND, MORE and MONEY for the assignment.
- Check whether $SEND + MORE$ equals MONEY.
- If it does, print the values of SEND, MORE and MONEY and stop.

Program

```
import itertools

letters = "SENDMORY"
digits = range(10)

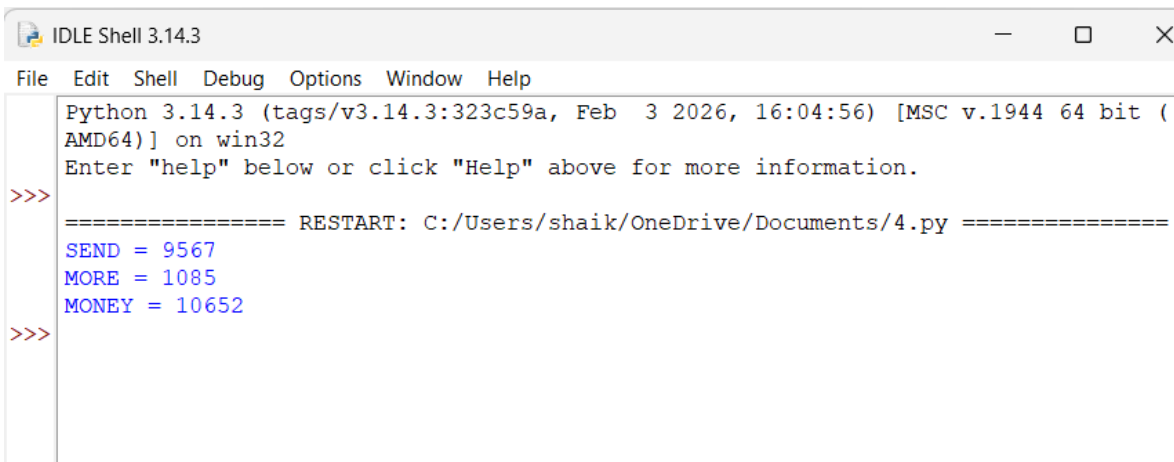
for p in itertools.permutations(digits, len(letters)):
    d = dict(zip(letters, p))

    if d["S"] == 0 or d["M"] == 0:
        continue

    send = 1000*d["S"] + 100*d["E"] + 10*d["N"] + d["D"]
    more = 1000*d["M"] + 100*d["O"] + 10*d["R"] + d["E"]
    money = 10000*d["M"] + 1000*d["O"] + 100*d["N"] + 10*d["E"] + d["Y"]

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Output



```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/shaik/OneDrive/Documents/4.py =====
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```

RESULT

Thus, the Crypt-Arithmetic Problem was solved successfully using Python.